

📌 How Many Ways Can We Create Beans in Spring?

- ♦ In Spring, a Bean is an object that is created, managed, and maintained by the Spring IoC Container.
- ♦ There are several ways to create beans in Spring.

1 Using @Component Annotation

- ♦ This is the most common approach.
- ♦ Spring automatically creates the bean during component scanning.

```
package com.vijaystack.service;

import org.springframework.stereotype.Component;

@Component
public class EmployeeService {
}
```

- ♦ Accessing the Bean

```
EmployeeService service =
    context.getBean(EmployeeService.class);
```

2 Using @Service Annotation

- ♦ @Service is a specialized version of @Component.
- ♦ It is used for Service Layer classes.

```
package com.vijaystack.service;

import org.springframework.stereotype.Service;

@Service
public class EmployeeService {
}
```

3 Using @Repository Annotation

- ♦ @Repository is used for DAO/Repository classes.

```
package com.vijaystack.repository;

import org.springframework.stereotype.Repository;

@Repository
public class EmployeeRepository {
}
```

4 Using @Controller Annotation

- ♦ Used to create Spring MVC Controller beans.

```
package com.vijaystack.controller;

import org.springframework.stereotype.Controller;

@Controller
public class EmployeeController {
}
```

5 Using @RestController Annotation

- ♦ Used to create REST API Controller beans.

```
package com.vijaystack.controller;

import org.springframework.web.bind.annotation.RestController;

@RestController
public class EmployeeRestController {
}
```

6 Using @Bean Annotation

- ♦ We can create beans explicitly inside a Configuration class.

```
package com.vijaystack.config;

import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;

@Configuration
public class AppConfig {

    @Bean
    public EmployeeService employeeService() {
        return new EmployeeService();
    }
}
```

- ♦ Accessing the Bean

```
EmployeeService service =
    context.getBean(EmployeeService.class);
```

7 Using XML Configuration

- ♦ Before Spring Boot, beans were commonly created using XML.

```
<bean id="employeeService"
      class="com.vijaystack.service.EmployeeService"/>
```

- ♦ Accessing the Bean

```
EmployeeService service =
    context.getBean("employeeService",
        EmployeeService.class);
```

Summary:

- ♦ Spring Bean Creation Methods:

👉 @Component

👉 @Service

👉 @Repository

👉 @Controller

👉 @RestController

👉 @Bean

👉 XML <bean> Tag

📌 In modern Spring Boot applications, the most commonly used approaches are:

✅ @Component

✅ @Service

✅ @Repository

✅ @Controller

✅ @RestController

✅ @Bean